

Week 4 Material - Inverse Trig, Tangent Lines and Linearization

15. Consider a ball that is being launched from the point $(0, 10)$, towards the right, at an angle α from horizontal. The ball will have an initial speed of 12 m/s. It will be subject to the force of gravity, but we will ignore air resistance.
- From intuition, what launch angle should lead to the ball going furthest?
 - Starting with the acceleration vector, find the position of the ball over time, $\mathbf{r}(t)$.
 - Find the horizontal distance traveled by the ball if it is launched at 45 degrees, and at 30 degrees. The ground is flat, at height $y = 0$.
 - Comment on your initial answer in part (a).

16. Find the exact value of the following expressions, without using a calculator.

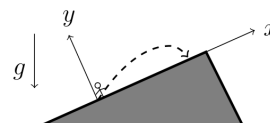
- $\arcsin\left(\sin\left(\frac{13\pi}{12}\right)\right)$.
- $x = \arccos\left(\cos\left(\frac{-5\pi}{8}\right)\right)$.
- $\cos\left(\arcsin\left(\frac{1}{2}\right)\right)$
- $y = \sin\left(\arctan\left(\frac{7}{2}\right)\right)$.
- $\sin(\arctan(x))$

17. Differentiate the following scalar and vector-valued functions.

- $f(s) = \frac{\arctan(s)}{4^s}$.
- $\vec{r}(t) = \left\langle \frac{\arccos(t)}{e^{2t}}, 10^{\sqrt{t}} \right\rangle$.

18. ⁶ You throw a ball up the side of a hill. Relative to the xy coordinate axes shown in the diagram to the right, the velocity of the ball at time t (in seconds) is given by

$$\vec{v}(t) = \left\langle \underbrace{-9t + 10}_{v_x(t)}, \underbrace{-9.8t + 6}_{v_y(t)} \right\rangle \text{ m/s.}$$



- Give a function $\vec{r}(t) = \langle x(t), y(t) \rangle$ which describes the position of the ball at time t . Assume that the ball is thrown from $\langle x, y \rangle = \langle 0, 0 \rangle$.
- When does the ball hit the hillside?
- Since there is no wind, the only acceleration on the ball is due to gravity. Give a vector in the hillside xy coordinates which points in the direction of gravity.
- The accelerations due to gravity on Earth, Mars and Neptune are 9.8 m/s^2 , 3.8 m/s^2 and 13.3 m/s^2 respectively. Which planet are we on? Be sure to justify your answer.

¹Source: previous APSC 171 Test 1

²Source: previous APSC 171 Test 1

³Source: previous APSC 171 Test 1

⁴Source: previous APSC 171 Test 1

⁵Source: previous APSC 171 Test 1

⁶Source: previous APSC 171 Test 1